

CourseCurator AI

The first adaptive learning platform
powered by Autonomous Agentic AI.

"We Don't Ask, We Derive."

HackArena 2.0 | Hyderabad Zonal Round Submission

01. The Broken EdTech Paradigm

Current platforms assume all students learn the same way. When they try to personalize, they rely on flawed self-reporting.

The Fallacy of Questionnaires

Every major app asks students: "What is your learning style?" Research shows students rarely know their true style. They guess, and the platform locks them into a sub-optimal path.

The Crisis in Numbers

- **67%**
of students report online courses don't match how they learn.
- **3**
sessions is all it takes for the average student to abandon a static course.
- **250M+**
students in India forced through one-size-fits-all formats.

02. The Paradigm Shift

We Don't Ask. We Derive.

Course Curator AI is driven by a passive, behavioral inference engine. We never ask a student how they learn. Our Agentic AI silently observes their natural interactions during their first video session.

Behavioral Signals

- **Rewinds & Pauses:** Indicates Auditory preference.
- **Panel Dwell Time:** Which generated content do they stare at?
- **Quiz Speed:** Active vs Reflective processing.

Instant Adaptation

- Detects Visual, Auditory, Read/Write, or Kinesthetic styles.
- Reorders the UI instantly to prioritize their preferred format.
- Achieved within just **45 seconds** of usage.

03. The Generative Payload

When a video opens, our Agentic AI autonomously orchestrates Google Gemini to generate 5 multimodal learning assets in parallel.



The Summary

For the **Read/Write Learner**. 5 structured, highly-condensed bullet points capturing core insights.



The Transcript

For the **Auditory Learner**. Timestamped, clickable lecture paragraphs that seek the video instantly.



The Mindmap

For the **Visual Learner**. An auto-generated SVG concept network linking ideas spatially.



The Flowchart

For the **Logical/Visual Learner**. Step-by-step process diagrams rendered live.



Personalized Quizzes

For the **Kinesthetic Learner**. Instantly generated MCQs with AI-powered explanations for every choice,



PPT Generation

For **Educators & Revision**. Auto-converts the topic structure into a downloadable presentation slide deck

04. Proactive Micro-Interventions

Course Curator AI doesn't just passively observe; it actively steps in when it detects friction, just like a human tutor.

Contextual Pause Analysis

When a student pauses the video, a tiny, non-intrusive micro-prompt asks: *"Taking notes 📝, Confused 😬, or Bored 🤔?"* We instantly map their exact intent to their learning profile rather than guessing.

Answering Style Evaluation

We track quiz response times and answer-changing behaviors to classify learners as **Reflective** (slow, accurate) or **Active** (fast, trial-and-error), adjusting the pace of future questions.

Dynamic Content Simplification

If the Agentic AI detects confusion (via "Confused" pauses or failed quizzes), it autonomously regenerates the summary using **"Explain Like I'm 5" (ELI5)** analogies to bridge the knowledge gap.

The Agentic Study Buddy

A proactive AI that intervenes right when needed: *"I noticed you struggled with the last concept. I just generated a simpler visual flowchart for that exact idea. Let's look at it."*

05. Generative vs. Agentic

Generative AI creates content. Agentic AI takes action. We use both.

The Autonomous Orchestrator

Most platforms use AI as a chatbot (Generative). We use AI as an orchestrator (Agentic).

- **Autonomous Trigger:** The moment a video loads, the agent fires 5 parallel prompts to Gemini. No user input required.
- **Self-Healing JSON:** The agent demands strict JSON schemas. If the LLM hallucinates formatting, the agent catches it and retries autonomously.
- **Fallback Strategy:** If API limits are hit, the agent gracefully degrades to rich, pre-built offline mocks—ensuring zero downtime during demos.
- **Profile Computation:** The agent acts as a silent observer, calculating real-time engagement vectors to update the student's Learning Profile.

06. Future Vision & Scaling

Where Agentic AI takes Course Curator next: pushing the boundaries of autonomous education.

The "Self-Healing" Curriculum

If a student fails a quiz, the AI identifies the missing prerequisite from a global knowledge graph and autonomously inserts a micro-lesson to bridge the gap.

Micro-Interaction Telemetry

Beyond pauses, we will monitor cursor movements and scroll velocity. Erratic movements indicate frustration, allowing for deeper, biometric-free behavioral profiling.

AI-Orchestrated Peer Learning

The AI compares learning profiles across users and autonomously pairs students with complementary strengths for a 5-minute collaborative "Boss Fight" challenge.

Cognitive Load Management

If quiz speed drops, indicating burnout, the AI enforces a gamified 5-minute Pomodoro break, generating a personalized AI meme related to the topic to reset attention.

07. Tech Stack & Impact

Technology Stack

- **Frontend:** Next.js 14, React, Tailwind-style Vanilla CSS.
- **AI Engine:** Google Gemini 1.5 Flash API.
- **State:** Zustand (Persistent Local Storage).
- **Visuals:** Custom interactive SVG renderers.

The Impact

- **Scale:** Targets 250M+ students in India.
- **Cost Efficiency:** ~₹0.02 API cost per video session.
- **Accessibility:** Transcripts aid hearing-impaired; visual maps aid dyslexic learners.
- **SDG Alignment:** UN Goal 4 — Quality Education.

Curator

Ready for the HackArena 2.0 In-Person Prototype Round.